



FAC & CoreHPC Demo

Presented by:

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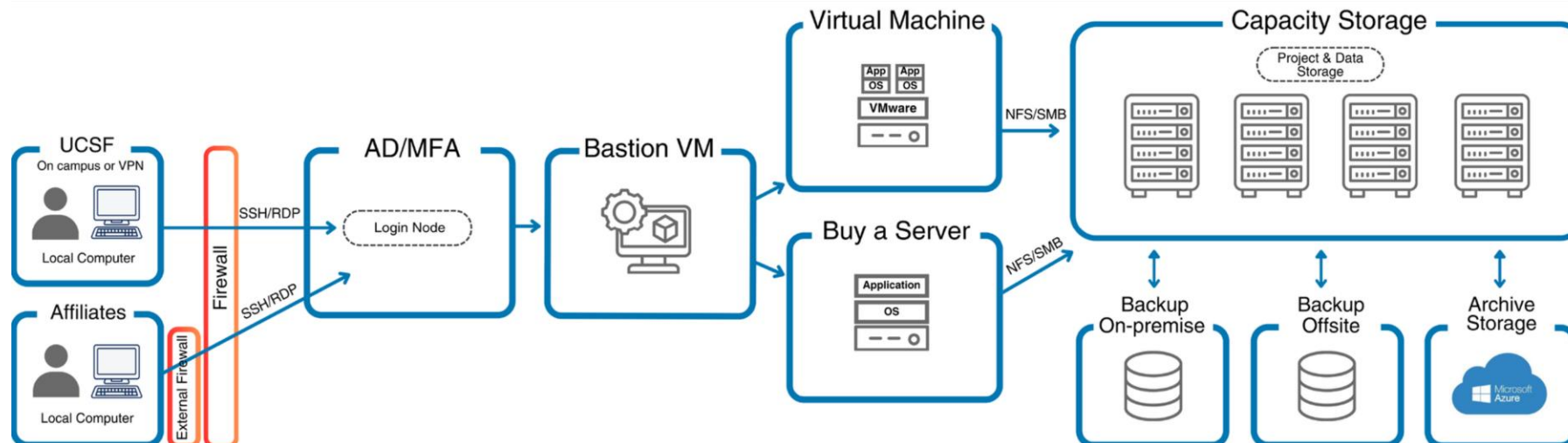
What will we learn?

- FAC & CoreHPC Overview and Concepts – **10 Minutes.**
- Access/Mount FAC Capacity Storage Drive - **10 Minutes**
 - MAC/Windows via SMB.
 - CoreHPC - via NFS Automounts
- Login to CoreHPC / Job Submission – **20 Minutes**
 - Access FAC Capacity Storage from CoreHPC
 - Copy from Capacity Storage to Fast Scratch
 - Submit SLURM Jobs.
 - Copy results from Fast Scratch to Capacity
- Question & Answers – **10 Minutes**

FAC Overview

What is FAC?

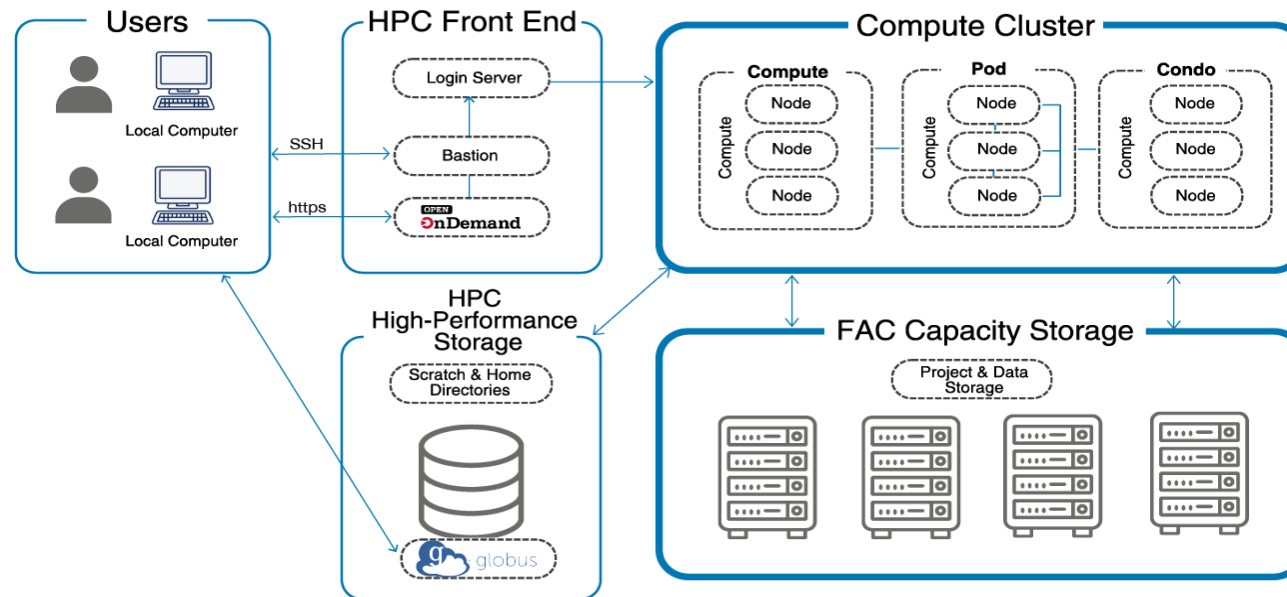
- The **Facility for Advanced Computing (FAC)** is a centralized service designed for **UCSF Researchers** for secure computing and capacity storage needs, managed by **UCSF Academic Research Services(ARS)**.
- It supports standalone computing
 - VM's and Physical servers deployed in dedicated Lab networks within FAC
 - Prototyping, interactive computing, website, database and custom application hosting.
- Capacity storage for large datasets
 - Central to all FAC/CoreHPC services and available on UCSF networks.
 - Backed up and Compliant with UCOP Policies (IS-3) and NIST 800-171



CoreHPC Overview

What is CoreHPC?

- **CoreHPC** is UCSF's high-performance computing cluster environment.
- Designed for scientific GPU/CPU & AI workloads.
- SLURM for batch job submission. Globus is available for data transfer.
- Thousands of CPU and GPU cores connected via InfiniBand & work together as cluster, connected to high-speed **Temporary Scratch Space**
- It is connected to **FAC Capacity Storage** on a high-speed network for **persistent** long-term storage.
- CoreHPC is available via yearly **user access fee** and a required **FAC Capacity storage** subscription.



FAC & CoreHPC Difference



FAC

Flexible computing & persistent storage



VMs

Web & database hosting, custom apps



Physical servers

Standalone, bare metal for prototyping



Capacity storage

Persistent Petabyte-scale warm Storage

Shared fast network

Capacity
Storage



CoreHPC

High-performance computing cluster



CPU & GPU clustered cores

Thousands of parallel cores



GPU AI Workloads

Ultra-low-latency interconnect



Scratch space & Slurm Scheduler

High-speed, short term for batch processing

Important Notes before we begin:

■ **Connection to FAC/CoreHPC:**

- Via UCSF Global Protect VPN.
- Access is provisioned via Campus or UCSFMC accounts, SOM accounts do not work with FAC.
- External users must affiliate with UCSF and have an UCSF account.

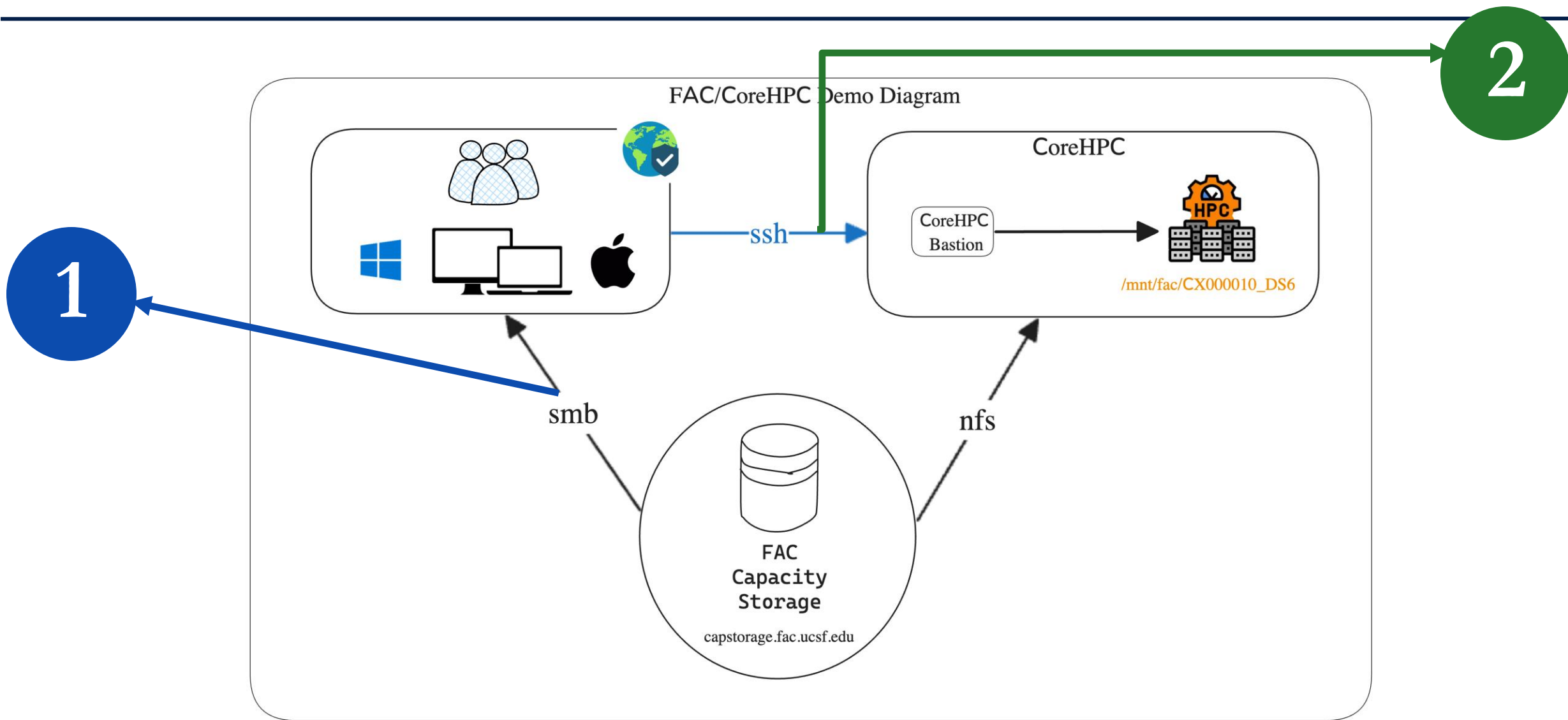
■ **FAC CX#:**

- Customer number generated for each LAB/Department/PI.

■ **FAC Datastore (DS)**

- Network shared drive on capacity storage for persistent/long-term data storage. **CX000000_DS1,2,3....**
- Accessed via workstations, FAC VM/Physical Servers/**CoreHPC**. Access is at data store level.
- A CX# can have multiple datastores for different projects, can be recharge separately.

Live Demo Session



Questions? & Quick Links

Our websites:

- ✓ [CoreHPC](#)
- ✓ [Facility for Advanced Computing\(FAC\)](#)
- ✓ [Wynton Transition](#)

Slide Decks

- ✓ [FAC Walking Deck](#)
- ✓ [CoreHPC Walking Deck](#)

Video References

- ✓ [FAC & CoreHPC Overview](#)
- ✓ [FAC & CoreHPC - How they work together](#)

Estimate Cost for FAC/CoreHPC:

- ✓ [FAC Cost Calculator](#)

Request FAC or CoreHPC

- ✓ [Request FAC](#)
- ✓ [Request CoreHPC](#)

Already onboarded and need FAC or CoreHPC support?

- ✓ [FAC Help](#)
- ✓ [CoreHPC Help](#)

ReadMe

- ✓ [CoreHPC Access Primer](#)
- ✓ [Mount FAC Capacity Storage](#)